

- Description

This module manages a vehicle fuel consumption monitoring system, ensuring that:

1. Fuel efficiency is calculated based on distance traveled and fuel used.
2. If fuel efficiency falls below a threshold, a low-efficiency warning is triggered.
3. The system suggests fuel-saving driving adjustments when efficiency is poor.
4. A log of fuel consumption, efficiency, and warnings is maintained for analysis.

This system helps improve fuel efficiency, reduce costs, and promote eco-friendly driving habits.

- Specification

```
module FuelEfficiencySystem
```

```
var distance, fuelUsed, efficiency, efficiencyThreshold: real
```

```
var consumptionLog: sequence of string
```

```
process recordTrip(distance: real, fuelUsed: real)
```

functional scenario form:

```
efficiency >= efficiencyThreshold && consumptionLog = consumptionLog ⊕
```

("Low Efficiency Warning: " + efficiency), Efficiency recorded successfully.

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$$\text{efficiency} < \text{efficiencyThreshold} \ \&\& \ \text{consumptionLog} = \text{consumptionLog} \oplus$$

("Efficiency Recorded: " + efficiency), Low efficiency warning triggered.

end process

end module

- Java Code

```
package code;
import java.util.*;
class FuelMonitor {
    private double efficiencyThreshold;
    private List<String> consumptionLog;
    public FuelMonitor(double efficiencyThreshold) {
        this.efficiencyThreshold = efficiencyThreshold;
        this.consumptionLog = new ArrayList<>();
    }
    public void recordTrip(double distance, double fuelUsed) {
        if (fuelUsed <= 0 || distance <= 0) {
            System.out.println("Invalid input: Distance and fuel consum
tion must be greater than zero.");
            return;
        }
        double efficiency = distance / fuelUsed; // km per liter
        String message = String.format("Distance: %.1f km | Fuel Used:
%.2f L | Efficiency: %.2f km/L",
distance, fuelUsed, efficiency);
```

```

        if (efficiency < efficiencyThreshold) {
            message += " | WARNING: Low Efficiency! Consider improving
driving habits.";
        }
        System.out.println(message);
        consumptionLog.add(message);
    }
    public void printConsumptionLog() {
        System.out.println("\nFuel Consumption Log:");
        for (String log : consumptionLog) {
            System.out.println(log);
        }
    }
}

public class c25_FuelEfficiencySystem {
    public static void main(String[] args) {
        FuelMonitor monitor = new FuelMonitor(15.0);
        // testcase-VT:
        //     monitor.recordTrip(-399.37,479.28);
        //     monitor.recordTrip(185.65,131.21);
        //     monitor.recordTrip(-237.52,-144.81);
        //     monitor.recordTrip(-276.69,480.34);
        //     monitor.recordTrip(430.27,-475.26);
        //     monitor.recordTrip(-185.11,323.64);
        //     monitor.recordTrip(336.08,61.85);
        //     monitor.recordTrip(81.00,262.77);
        //     monitor.recordTrip(423.02,-53.50);
        //     monitor.recordTrip(22.17,129.93);
        //     monitor.recordTrip(-258.12,430.96);
        //     monitor.recordTrip(-320.86,159.80);
        //     monitor.recordTrip(-95.79,456.37);
        //     monitor.recordTrip(-361.48,-251.44);
        //     monitor.recordTrip(305.07,483.99);
        //     monitor.recordTrip(-408.17,-222.20);
        //     monitor.recordTrip(-274.78,-453.53);
        //     monitor.recordTrip(193.13,243.25);
        //     monitor.recordTrip(-200.49,397.34);
        //     monitor.recordTrip(-356.27,120.59);
        //     monitor.recordTrip(88.77,253.01);
        //     monitor.recordTrip(-211.50,-45.15);
        //     monitor.recordTrip(-332.76,454.70);
        //     monitor.recordTrip(-471.61,-251.61);
        //     monitor.recordTrip(-448.71,20.88);
        //     monitor.recordTrip(215.90,-227.64);
    }
}

```

```
//      monitor.recordTrip(-307.14, -289.79);
//      monitor.recordTrip(250.62, 471.65);
//      monitor.recordTrip(-81.10, -464.02);
//      monitor.recordTrip(139.46, 58.82);
//      monitor.recordTrip(293.08, 388.58);
//      monitor.recordTrip(60.36, -221.05);
//      monitor.recordTrip(363.82, 78.01);
//      monitor.recordTrip(449.93, -411.57);
//      monitor.recordTrip(-497.38, 168.99);
//      monitor.recordTrip(-275.05, -30.13);
//      monitor.recordTrip(182.19, -32.25);
//      monitor.recordTrip(-340.23, 25.91);
//      monitor.recordTrip(-134.46, -310.97);
//      monitor.recordTrip(-443.19, 240.56);
//      monitor.recordTrip(-356.40, 214.22);
//      monitor.recordTrip(240.24, 298.57);
//      monitor.recordTrip(-65.38, -71.44);
//      monitor.recordTrip(-331.92, -109.53);
//      monitor.recordTrip(-377.65, 225.58);
//      monitor.recordTrip(-497.41, -127.34);
//      monitor.recordTrip(35.23, 441.17);
//      monitor.recordTrip(-330.93, 471.64);
//      monitor.recordTrip(234.73, 389.15);
//      monitor.recordTrip(333.65, 85.61);
//      monitor.recordTrip(175.99, -240.40);
//      monitor.recordTrip(441.78, -395.32);
//      monitor.recordTrip(159.47, -447.02);
//      monitor.recordTrip(-61.96, -412.06);
//      monitor.recordTrip(187.01, -170.70);
//      monitor.recordTrip(375.89, 175.00);
//      monitor.recordTrip(394.18, -468.78);
//      monitor.recordTrip(487.58, -75.78);
//      monitor.recordTrip(273.64, 444.20);
//      monitor.recordTrip(189.96, 305.75);
//      monitor.recordTrip(311.52, 424.12);
//      monitor.recordTrip(274.59, 136.11);
//      monitor.recordTrip(-125.33, 2.10);
//      monitor.recordTrip(427.86, 247.89);
//      monitor.recordTrip(464.34, -392.17);
//      monitor.recordTrip(-273.54, 435.47);
//      monitor.recordTrip(-6.43, 356.74);
//      monitor.recordTrip(-454.04, 308.86);
//      monitor.recordTrip(215.02, 170.34);
//      monitor.recordTrip(30.43, -4.12);
```

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

```
//      monitor.recordTrip(-340.23,25.91);
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//      monitor.recordTrip(-454.04,308.86);
//      monitor.recordTrip(215.02,170.34);
//      monitor.recordTrip(30.43, -4.12);
//      monitor.recordTrip(-442.96,394.63);
//      monitor.recordTrip(-36.30,386.29);
//      monitor.recordTrip(39.82, -92.78);
//      monitor.recordTrip(245.87,355.27);
//      monitor.recordTrip(-299.50, -466.62);
//      monitor.recordTrip(-464.10, -63.60);
//      monitor.recordTrip(188.00,58.41);
//      monitor.recordTrip(407.45, -164.48);
//      monitor.recordTrip(184.62, -336.48);
//      monitor.recordTrip(-155.23,483.90);
//      monitor.recordTrip(-359.17, -302.07);
```

```
//      monitor.recordTrip(-114.45,277.38);
//      monitor.recordTrip(-414.38,-473.13);
//      monitor.recordTrip(368.83,90.54);
//      monitor.recordTrip(291.62,-467.71);
//      monitor.recordTrip(264.90,-392.46);
//      monitor.recordTrip(278.58,364.75);
//      monitor.recordTrip(-43.22,81.87);
//      monitor.recordTrip(211.04,-100.30);
//      monitor.recordTrip(-356.88,-363.49);
//      monitor.recordTrip(454.10,341.97);
//      monitor.recordTrip(327.98,-94.73);
//      monitor.recordTrip(35.35,408.34);
//      monitor.recordTrip(437.67,-209.76);
//      monitor.recordTrip(-144.69,407.91);
//      monitor.recordTrip(323.14,-489.44);
//      monitor.recordTrip(194.79,-1.91);
//      monitor.recordTrip(-457.85,286.00);
//      monitor.recordTrip(-112.15,-480.38);
//      monitor.recordTrip(-23.12,-208.37);
```

```
// testcase-UVT:
```

```
monitor.recordTrip(-399.37,479.28);
monitor.recordTrip(185.65,131.21);
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monitor.recordTrip(-211.50,-45.15);
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```

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        monitor.recordTrip(-457.85,286.00);
        monitor.recordTrip(-112.15,-480.38);
        monitor.recordTrip(-23.12,-208.37);

        monitor.printConsumptionLog();
    }
}

```

- Coverage report

1. Test case number: 100
2. V-Method testing branch coverage: 87.5%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.8 %	7	835	842
▼ src	0.8 %	7	835	842
▼ code	0.8 %	7	835	842
▼ c25_FuelEfficiencySystem.java	87.5 %	7	1	8
> FuelMonitor	87.5 %	7	1	8
> c25_FuelEfficiencySystem		0	0	0

3. Vibration testing instruction coverage: 99.4%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	2.9 %	491	16,712	17,203
▼ src	2.9 %	491	16,712	17,203
▼ code	2.9 %	491	16,712	17,203
▼ c25_FuelEfficiencySystem.java	99.4 %	491	3	494
> c25_FuelEfficiencySystem	99.3 %	408	3	411
> FuelMonitor	100.0 %	83	0	83

4. Fuzz testing branch coverage: 75%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.7 %	6	836	842
▼ src	0.7 %	6	836	842
▼ code	0.7 %	6	836	842
▼ c25_FuelEfficiencySystem.java	75.0 %	6	2	8
> FuelMonitor	75.0 %	6	2	8
> c25_FuelEfficiencySystem		0	0	0

5. Fuzz testing instruction coverage: 99.4%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	2.9 %	491	16,718	17,209
▼ src	2.9 %	491	16,718	17,209
▼ code	2.9 %	491	16,718	17,209
▼ c25_FuelEfficiencySystem.java	99.4 %	491	3	494
> c25_FuelEfficiencySystem	99.3 %	408	3	411
> FuelMonitor	100.0 %	83	0	83

6. Z3 branch coverage: 62.5%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.6 %	5	835	840
▼ src	0.6 %	5	835	840
▼ code	0.6 %	5	835	840
▼ c25_FuelEfficiencySystem.java	62.5 %	5	3	8
> FuelMonitor	62.5 %	5	3	8
> c25_FuelEfficiencySystem		0	0	0

7. Z3 instruction coverage: 95.5%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	2.1 %	470	21,628	22,098
▼ src	2.1 %	470	21,628	22,098
▼ code	2.1 %	470	21,628	22,098
▼ c25_FuelEfficiencySystem.java	95.5 %	470	22	492
> FuelMonitor	77.1 %	64	19	83
> c25_FuelEfficiencySystem	99.3 %	406	3	409

8. Unimproved V-method branch coverage: 62.5%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.6 %	5	835	840
▼ src	0.6 %	5	835	840
▼ code	0.6 %	5	835	840
▼ c25_FuelEfficiencySystem.java	62.5 %	5	3	8
> FuelMonitor	62.5 %	5	3	8
> c25_FuelEfficiencySystem		0	0	0

9. Unimproved V-method instruction coverage: 95.5%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
VibrationTestingTool	2.1 %	470	21,628	22,098
src	2.1 %	470	21,628	22,098
code	2.1 %	470	21,628	22,098
c25_FuelEfficiencySystem.java	95.5 %	470	22	492
FuelMonitor	77.1 %	64	19	83
c25_FuelEfficiencySystem	99.3 %	406	3	409